

THE ABSOLUTE SENSITIVITY OF RADIO RECEIVERS

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Summary—The total random noise originating in a receiver has customarily been described in terms of the equivalent noise voltage at the receiver input terminals. A comparison of the signal-to-noise ratios of two receivers working out of identical antennas is thereby facilitated, but only so long as the coupling between antenna and receiver input is extremely loose.

This paper describes a method for rating and measuring the noise in complete receiving systems, antenna included. The proposed rating appears particularly applicable to ultra-high-frequency services and, more generally, to any service in which signal-to-noise ratio is made a prime consideration in receiver design and operation.

A portion of the study deals with the properties of receiving antennas, yielding as a by-product an alternative derivation of Nyquist's theorem concerning thermal fluctuations in passive networks.

A formula for absolute sensitivity is developed, which shows how the minimum usable signal-field strength is related to the operating wavelength, the antenna directivity, the local noise-field strength, the receiver bandwidth, and a number called the "noise factor", which is a basic measure of the internal noise sources of the receiver.

THE "Standards on Radio Receivers" adopted and published by the Institute of Radio Engineers in 1938, define the sensitivity of a radio receiver as "that characteristic which determines the minimum strength of signal input capable of causing a desired value of signal output".¹ So defined, the sensitivity is a measure of gain. The same report² takes cognizance of the fact that random fluctuations (noise) originating in the receiver set a limit to the useful sensitivity. Leaving a precise definition to those who write standards, we shall refer to this useful limit as the "absolute sensitivity", indicate in an elementary way its dependence upon certain properties of the receiver³ and antenna, suggest methods for measuring it in the laboratory, and consider its modification by noise induced in the antenna.

¹ "Standards on Radio Receivers", 1938, definition 1R36.

² Page 42, section 13.

³ A more thorough consideration of this aspect is the purpose of a companion paper by E. W. Herold, "An Analysis of the Signal-to-Noise Ratio of Ultra-High-Frequency Receivers", RCA REVIEW, Jan. 1942.

The conventional method of describing receiver noise in terms of an equivalent-noise-side-band input, while possessing a limited utility, does not ever permit an immediate positive judgment of signal-to-noise ratio, nor even a comparison of two receivers, unless additional information is given concerning the antennas and the means for coupling. It will be seen, on the other hand, that a description of the absolute sensitivity of a receiver, antenna included, provides a direct basis for the judgment of a complete receiving system, and facilitates the inter-comparison of receiving equipments, no matter how diverse their characteristics.⁴ A rating of this kind is particularly valuable in the ultra-high-frequency field, where antennas are designed as an integral part of the receiver, or, for that matter, in any service wherein maximum signal-to-noise ratio is an important factor in design and operation. It was during an attempt, two years ago, to find a sound basis for the comparison of television receivers, that the considerations reported below were crystallized.

MEASUREMENTS IN THE LABORATORY

Consider an ultra-high-frequency receiver constructed to work out of a specific antenna. This implies a specific radiation resistance, antenna reactance, and ohmic loss (such as might be present in a long down-lead). The important essentials of a laboratory simulation of the real antenna, therefore, appear to be

1—a signal generator providing a known radio-frequency voltage source, e , in series with

2—a resistor, R_a , whose resistance equals the radiation resistance of the prescribed real antenna;

3—tunable reactance to simulate the Q of the real antenna; perhaps

4—additional losses to simulate unavoidable losses in the real antenna, and possibly

5—a low-loss transformer to correct *minor* defects in the attempted similitude.

In many cases, the real antenna is prescribed to have a low Q , to permit reception over a wide range of frequencies. In this event, and provided the antenna Q is small in comparison with the Q of the receiver input circuit, item (3) is of minor importance. By the same token, emphasis upon "low-loss" in item (5) becomes less severe. Furthermore, item (5) will often be found incorporated in the receiver proper, in which case the responsibility for making it loss-free lies with the receiver designer.

⁴ This same subject has recently received the capable attention of R. E. Burgess, "Noise in Receiving Aerial Systems", *Proc. Phys. Soc.*, Vol. 53, p. 293, May, (1941).

Consider next methods for measuring the noise output. Present I.R.E. standards prescribe measurement of noise after detection.² This method not only requires a known modulation for the signal generator, but also makes the noise measurement a function of the frequency response of the circuits which follow the detector. On the premise that post-detection filters are, or should be made sufficiently flat and broad to pass uniformly all frequencies offered by the pre-detection part of the receiver, it is here proposed to measure noise prior to detection. The natural point at which to insert a meter is just ahead of the detector. But occasionally latitude is permissible, and the demands made upon the meter and its position may thus be summarized.

1—It must be preceded by all the filters which materially determine the r-f and i-f selectivity, and must not, in itself, modify the selectivity.

2—It must be preceded by all of the significant sources of random fluctuations.

3—Its response to both noise and signal (preferably, to any admixture of these) must be known.

Item (1) is often too stringent, and may be relaxed in circumstances where it is known that the frequency spectrum of the noise power presented to the meter input terminals conforms essentially to the shape of the power-selectivity curve at that point. This point will be elaborated later.

A schematic of such a laboratory arrangement is shown in Figure 1, which, in addition displays *one* of the sources of noise, namely, thermal agitation in the resistor R_a which, at room temperature T_o , is simulating radiation resistance. The quantity k appearing in the formula,

$$\overline{e_i^2} = 4k T_o R_a \Delta f \quad (1)$$

is Boltzmann's constant. The quantity Δf we will consider related to the overall selectivity curve in the usual way. That is, if f_o be the nominal operating frequency of the receiver, and if $G(f)$ be the response of a linear output meter to input signal $e(f)$, then

$$\Delta f = \int \frac{G^2(f)}{G^2(f_o)} df \quad (2)$$

By any of various methods which avoid overloading any portion of the system, it is possible to find a signal $e(f_o)$ which produces (as indicated by the meter) an output signal power equal in magnitude to the output noise power. Were the receiver ideal in the sense that it possessed no sources of noise save that originating in the dummy

² loc. cit.

antenna, it would be found that

$$\overline{e^2}(f_o) = 4k T_o R_a \Delta f \quad (3)$$

A fundamental description of the noise generated in the laboratory by a receiver equipped with a dummy antenna is, therefore, provided by the dimensionless measure

$$N = \frac{\overline{e^2}(f_o)}{4k T_o R_a \Delta f} \quad (4)$$

The number N we shall refer to as the "noise-factor". Expressed as a number, it states the ratio of actual noise power to that developed by an ideal receiver. Or, it may, of course, be given in decibels referred to the ideal receiver as a zero level. That is, the lower limit of N is unity, or zero decibels.

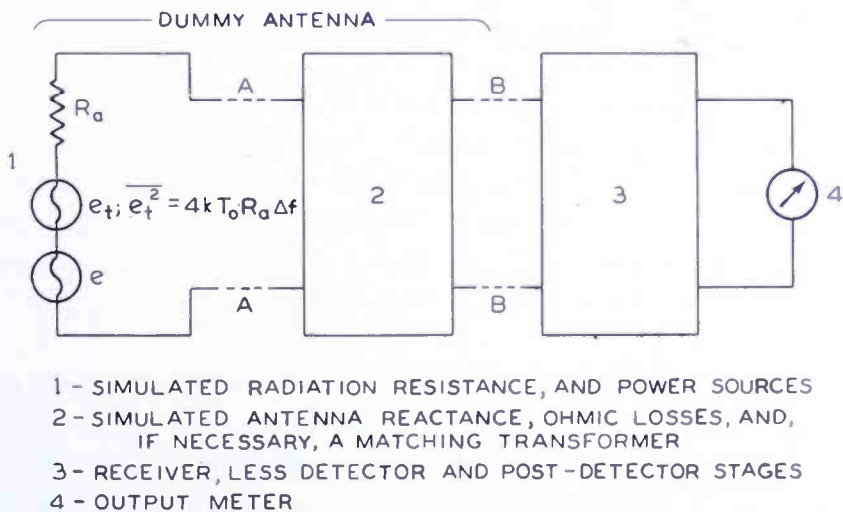


Fig. 1—Schematic of a receiving system with dummy antenna and output meter for laboratory measurements.

This proposed method for quoting laboratory measurement of receiver noise is particularly valuable for two reasons.

First, it is generally fairly insensitive to modifications in the selectivity, as was hinted above. For, unless the frequency spectrum of the noise power output differs materially from the power-selectivity curve, $\overline{e^2}(f_o)$ is proportional to Δf . The noise factor N thus provides a reasonably just basis for the comparison of equipments possessing selectivity curves widely different as to both shape and width.

Second, the quantity e^2/R_a is invariant in a transformation through a loss-free transformer. Consequently, whenever there are no prescribed ohmic losses in the antenna, and when the object of the measurement is not simply to measure N , but rather, to adjust the antenna coupling to an optimum which then produces the *lowest pos-*

sible N , one need not be limited to a value of R_a precisely equal to the radiation resistance of the prescribed antenna. One need only make sure that the prescribed antenna Q is provided, and even this requirement loses importance when the Q is too low to affect materially the overall selectivity. For all special, but highly significant measurements of this kind, it becomes unnecessary to know either the voltage calibration of the signal generator, or the precise value of R_a . Since only the quantity $\overline{e^2/R_a}$ appears in the expression for the noise factor, it is sufficient to know simply the power the combination will deliver to a matched load. In view of this property, it is seen that the minimum noise factor provides, also, a reasonably just basis for the comparison of equipments working out of antennas possessing widely different radiation resistances.

SOME PROPERTIES OF RECEIVING ANTENNAS

Although we have obtained a method for determining and rating the noise of a receiver with dummy antenna, we are not yet in a position to describe its absolute sensitivity in operation. We must first learn two things: one, the connection between signal-field strength and voltage e produced thereby at the open terminals of an antenna of radiation resistance R_a ; two, the rôle played by noise picked up by the antenna.

The rôle played by antenna reactance is altogether silent and will be ignored.

Consider two antennas as pictured in Figure 2. The reciprocity law (which is only a dignified way of saying that transfer impedances for passive networks are the same in both directions) assures us that if i is the current fed to one antenna, and e the resulting open-circuit voltage at the other, then

$$e = k i$$

and k is symmetrical in the two antennas. But if antenna (b) is transmitting, its power output is

$$P = i^2 R_b$$

And if antenna (a) is receiving some of that power, its open-circuit voltage will be proportional to the square-root of P . Therefore,

$$e \propto \sqrt{P} = \sqrt{R_b} i$$

It follows that k is proportional to $\sqrt{R_b}$, and, being symmetrical in the two antennas, must be proportional to $\sqrt{R_a}$ also. It is, in addition, proportional to a space-attenuation factor and to the product of the directivities of the two antennas.

We are thus enabled to write the relation between signal field strength and open-circuit antenna voltage. Let Ω represent the two angular coordinates which define a direction in space. Let ϕ represent the azimuth in a normal plane, to define the direction of polarization of a Poynting vector $S(\Omega, \phi)$ which describes the signal field. Let $D^2(\Omega, \phi)$ be the function which describes the power-directivity receiving pattern (identical with the radiation pattern in unencumbered

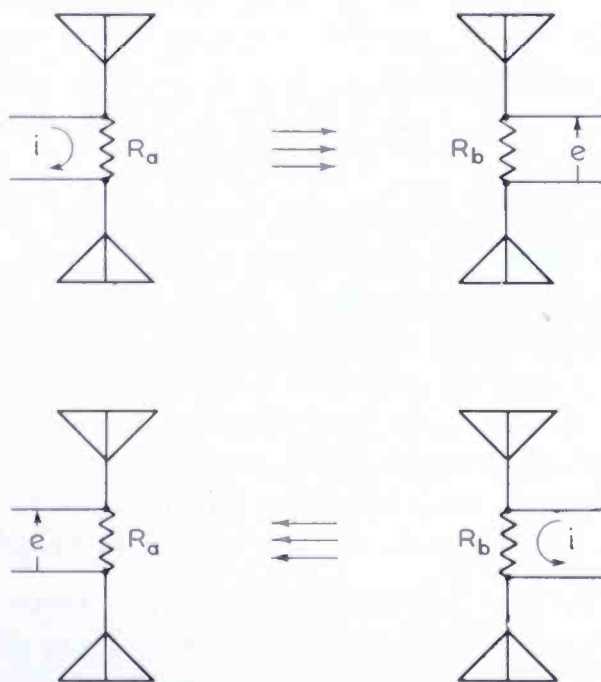


Fig. 2—The reciprocity law for two antennas with individual radiation resistances R_a and R_b .

space), and let it be normalized so that, averaged over all values of Ω and ϕ , $\overline{D^2(\Omega, \phi)} = 1$. It follows that

$$e^2 = A R_a D^2(\Omega, \phi) S(\Omega, \phi) \quad (5)$$

The coefficient A is evaluated in the Appendix by an appeal to Nyquist's theorem, for which an additional derivation is produced, proving (if, indeed, proof be needed) its valid application to radiation resistance as well as ohmic resistance. The result, Equation (15), is

$$A = \lambda^2 / 2\pi$$

where λ is the wavelength.

$$\text{Hence,} \quad e^2 = \frac{\lambda^2}{2\pi} R_a D^2(\Omega, \phi) S(\Omega, \phi) \quad (6)$$

When practical units are used, and λ is given in meters, the relation between S (watts per square meter) and its associated field strength E (volts per meter) is

$$S = E^2 / 120\pi \quad (7)$$

The open-circuit voltage produced at the antenna terminals by a signal field is then

$$e^2 = \frac{\lambda^2}{2\pi} \frac{R_a}{120\pi} D^2(\Omega, \phi) E^2 \quad (8)$$

The voltage produced by a noise field may be likewise expressed. It is certain that if the antenna were exposed solely to the isotropic equilibrium-radiation of an enclosure maintained at room temperature T_o , the noise voltage would be

$$\overline{e_t^2} = 4k T_o R_a \Delta f$$

It is equally certain that this is not the case. The strength of local noise fields has been, and will always continue to be, the subject of much serious speculation and measurement.⁵ While it is known already that both man-made noise and atmospherics decrease rapidly with λ , the "resolving power" of receivers is not yet sufficient to determine what limit, if any, is approached. Furthermore, the development of new communications equipment for shorter wavelengths will always, of itself, implement the production of what must necessarily be regarded, from certain points of view, as new noise fields.

Local noise fields may be crudely divided into three groups:

1—Strictly random fluctuations similar to those generated within the receiver proper, and characterized by a proportionality between noise power and bandwidth.

2—Noise consisting of impulses, random enough, but occurring at a mean rate too low in comparison with the bandwidth to be included with (1), and, therefore, characterized by a lack of proportionality between noise power and bandwidth.

3—All other unwanted fields.

It is beyond our present purposes to discuss the relative amounts of service disturbance engendered by equal amounts of noise power in each of these categories. This vastly complex problem is receiving attention elsewhere.⁶ Concerning ourselves with a measure of the

⁵ For example, K. G. Jansky, "Minimum Noise Levels Obtained on Short-Wave Radio Receiving Systems", *Proc. I.R.E.*, Vol. 25, p. 1517, December, (1937).

R. K. Potter, "An Estimate of the Frequency Distribution of Atmospheric Noise", *Proc. I.R.E.*, Vol. 20, p. 1512, September, (1932).

G. Reber, "Cosmic Static", *Proc. I.R.E.*, Vol. 28, p. 68, February, (1940).

⁶ C. M. Burrill, "Progress in the Development of Instruments for Measuring Radio Noise", *Proc. I.R.E.*, Vol. 29, p. 433, August, (1941).

power alone, we note that the difference between two determinations of the noise factor, conducted as described, one with a dummy antenna, the other with the real antenna in the absence of a signal field, provides just such a measure. We may go even further and, simply for convenience, express this measure in terms of an essentially fictitious temperature T_a of local space, writing

$$\overline{e^2} = 4k T_a R_a \Delta f$$

for the net open-circuit voltage produced by the local noise field at the antenna terminals, so that⁷

$$N(\text{operating}) = N + \frac{T_a}{T_o} - 1 \quad (9)$$

The total equivalent noise voltage at the antenna terminals of a system in operation may consequently be written

$$\overline{e^2} = 4k T_o R_a \Delta f \left[N + \frac{T_a}{T_o} - 1 \right] \quad (10)$$

Should it be ascertained that the local noise field consists entirely of the first kind, T_a will then be independent of bandwidth, but not necessarily independent of operating frequency; and if the noise field is not isotropic, T_a will naturally be a function of antenna orientation and directivity. To the extent that noise fields of the second and third kinds are present, T_a cannot even be presupposed independent of bandwidth.

ABSOLUTE SENSITIVITY

If, by the term "absolute sensitivity", we refer to the r-m-s signal field strength which produces at the open antenna terminals a mean-square voltage equal to the equivalent mean-square noise voltage, then, from (8) and (10), that field strength is

$$\overline{E^2} = \frac{240\pi^2}{\lambda^2} \cdot \frac{4k T_o \Delta f}{D^2(\Omega, \phi)} \left[N + \frac{T_a}{T_o} - 1 \right] \quad (11)$$

⁷ E. W. Herold, Reference 3, exhibits the general functional relationship of N to the important sources of noise *within the receiver*, and to the circuit arrangements dictated by the service for which it is designed.

If E is conventionally expressed in *microvolts per meter*, λ in meters, Δf in megacycles, and T_o set equal to 300 degrees Kelvin (room temperature),

$$\overline{E^2} = \frac{39\Delta f}{\lambda^2 D^2(\Omega, \phi)} \left[N + \frac{T_a}{300} - 1 \right] \quad (12)$$

The expression shows clearly the distinct, prominent contributions of receiver proper, antenna directivity, and local noise fields.

The economy of efforts to improve the laboratory noise factor is seen to be bounded, in some cases sharply, by the existence of local noise. The use of tuned antennas for broadcast reception is, in many locations, a costly luxury. In some recent experiments, W. R. Ferris⁸ and the author produced a noise factor $N = 3.2$ in an attempt to determine the lower limit for television service at 100 megacycles. Such a receiver would probably be unappreciated in metropolitan districts. On the other hand, at higher frequencies, T_a appears to decline, as stated before, while minimum N rises rapidly for a number of tube and circuit reasons.³ There is surely much to be gained from improved N 's at very short wavelengths.

The economy of antenna structures needs similar consideration. For half-wave dipoles in free space, D^2 is not a function of wavelength. For dipoles beamed by parabolas large in comparison with λ , D^2 (maximum) is proportional to A/λ^2 , where A is the area of the parabola's aperture. Even were the noise factor to remain fixed as one moved towards shorter wavelengths, the absolute sensitivity would certainly suffer if one continued to use dipoles and would, indeed, still suffer, despite the switch to a parabola, unless the aperture of the parabola were given an area comparable to the square of the initial wavelength.

CONCLUSION

Because of the ease of interpretation, and the simplicity of measurement method, it is hoped that the material presented above may be considered as a basis for the adoption of a standard of "absolute sensitivity" for radio receivers.

APPENDIX

Thermodynamic reasoning alone is sufficient to justify the application of Nyquist's theorem⁹ concerning thermal agitation to any

³ loc. cit.

⁸ Research Laboratories, RCA Manufacturing Company, Inc., Harrison, New Jersey.

⁹ H. Nyquist, "Thermal Agitation of Electric Charge in Conductors", *Phys. Rev.*, Vol. 32, p. 110, July, (1928).

passive network. But it is, at least, interesting to develop the theorem in detail for a microscopic dipole exposed to equilibrium radiation.¹⁰

Using electrostatic units, we know that the radiation resistance of a dipole of length l ($l \ll \lambda$) in vacuum is¹¹

$$R_a = \frac{8}{3} \frac{(\pi f l)^2}{C^3} \quad (13)$$

where f is the operating frequency, C the velocity of light.

The energy density of equilibrium radiation at temperature T is¹²

$$\rho = \frac{8\pi h f^3 df}{C^3} \left[e^{\frac{hf}{kT}} - 1 \right]^{-1}$$

wherein h is Planck's constant, k is Boltzmann's constant. But since $hf/kT \sim 10^{-4}$ for $f = 1,000$ megacycles and $T = 300$ degrees Kelvin, the Rayleigh-Jeans approximation is sufficiently accurate. Thus,

$$\rho = \frac{8\pi k T f^2}{C^3} df$$

But, being isotropic,

$$\rho = \frac{\overline{E^2}}{4\pi} = \frac{\overline{E_x^2} + \overline{E_y^2} + \overline{E_z^2}}{4\pi} = \frac{3\overline{E_z^2}}{4\pi}$$

Hence, the mean-square field strength at the antenna in its susceptible direction (z) is

$$\overline{E_z^2} = \frac{32(\pi f)^2}{3 C^3} kT df$$

Therefore, the mean-square open-circuit voltage at the antenna terminals is, with the help of (13),

$$\overline{e^2} = \overline{E_z^2} l^2 = 4 \left[\frac{8(\pi f l)^2}{3 C^3} \right] kT df = 4kT R_a df \quad (14)$$

* * * * *

To evaluate the coefficient A in (5), we submerge any antenna in equilibrium radiation and demand that the open-circuit voltage appear-

¹⁰ A more general derivation was given by Burgess, Reference 4.

¹¹ For example, Hund, "High-Frequency Measurements", p. 399, (McGraw-Hill, 1933).

¹² For example, Ruark and Urey, "Atoms, Molecules, and Quanta", p. 58, Eq. 10, (McGraw-Hill, 1930).

ing at its terminals obey Nyquist's theorem. Since $D^2(\Omega, \phi)$ has been defined to have an average value of unity, and since (in *e.s.u.*)

$$\bar{S} = -\frac{C \overline{E^2}}{4\pi}$$

it follows that

$$4kT R_a df = \overline{e^2} = A R_a \bar{S} = A R_a \frac{C \overline{E^2}}{4\pi} = \frac{2\pi A}{\lambda^2} \cdot 4kT R_a df$$

whence

$$A = \frac{\lambda^2}{2\pi} \quad (15)$$

This result enables one to state a perfectly general relationship between the "effective height" of an antenna, its radiation resistance, and its directivity. The effective height $h(\Omega, \phi)$ is defined

$$h = \frac{e}{E}$$

Hence, from (8), we have, in practical units,¹³

$$h^2(\Omega, \phi) = \frac{\lambda^2}{2\pi} \frac{R_a}{120\pi} D^2(\Omega, \phi) \quad (16)$$

We further find, in agreement with Burgess,⁴ that the mean-square effective height averaged over all orientations with respect to a fixed signal vector is then

$$\overline{h^2} = \frac{\lambda^2}{2\pi} \frac{R_a}{120\pi} \quad (17)$$

¹³ The quantity, 120π , carries the dimension, ohms. It is, in fact, the characteristic impedance of a strip x units broad of a transmission line consisting of two parallel infinite planes x units apart, supporting plane-wave transmission.

⁴ loc. cit.

Finally, with reference to (6), it is to be noticed that the power an antenna can extract from a signal Poynting vector is proportional to

$$\frac{e^2}{R_a} = \frac{\lambda^2}{2\pi} D^2(\Omega, \phi) S(\Omega, \phi)$$

The radiation pattern of dipoles shorter than a half-wavelength is sensibly independent of the length; this leads to the curious observation that such dipoles extract sensibly equal amounts of power, no matter how short they are. In the language of physical optics this would be known as a diffraction phenomenon. In the language of atomic physics, we can only conclude that the capture-cross-section of short dipoles for passing Poynting vectors is of the order of λ^2 .